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RESEARCH ARTICLE

# Federated Learning for Privacy-Preserving Employee Performance Analytics

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**ABSTRACT** With the increasing sensitivity surrounding employee performance data, there is a pressing need for predictive systems that preserve privacy while delivering actionable insights to organizations. This paper introduces HFAN-Priv, a hierarchical federated attention network designed to predict employee resignation risk and evaluate performance trends without sharing raw data across organizations. The framework integrates feature-level and instance-level attention to model complex workforce patterns, applies differential privacy through gradient masking to ensure compliance with data protection regulations, and enhances interpretability using local SHAP and LIME explanations. Experiments conducted on a real-world employee productivity dataset show that HFAN-Priv achieves near-perfect predictive accuracy, robust cross-client generalization, and transparent decision-making, all while maintaining strong privacy guarantees. The proposed approach presents a scalable, ethical, and effective solution for HR analytics in decentralized environments.

**INDEX TERMS** Federated learning, employee performance, privacy-preserving AI, differential privacy, SHAP, human resource analytics.

## I. INTRODUCTION

Employee performance analytics is becoming central to the way organizations manage their workforce [1]. Human resource teams now collect large amounts of data about employees. This includes data on productivity, attendance, satisfaction, and behavior [2]. The goal is to understand what leads to good performance or early resignation. Using this data, machine learning models can predict who might leave, who deserves a promotion, or who may need support [3]. These models help companies plan better and reduce risk. However, employee data is highly sensitive. It contains private information about salaries, medical leaves, performance scores, and personal demographics [4]. This creates a challenge. Many companies are unable to use centralized learning systems because those systems require moving raw data into a single location. Such data movement is often not allowed due to internal rules or privacy regulations [5].

Organizations must find a way to use data while protecting privacy. Most HR records are not just about performance [6]. They also include sensitive details like resignation status, team size, training hours, and education background [7]. If a breach happens, it can cause reputational damage and legal problems. As a result, many companies avoid data sharing. They limit model training to single systems. This means they miss out on learning across teams or offices. A company may have the same problem in many branches but still fail to connect the dots because of these data walls [8].

Despite these concerns, companies still need to find patterns across large systems. Human resource teams want to see how performance trends differ by region. They want to learn whether certain training programs are working. They want to detect if some job roles are at higher risk of resignation. But these questions are hard to answer with isolated data. Most systems are not built to share insights across locations. Even when branches want to collaborate, they cannot transfer raw records due to strict rules [9]. The challenge is to make models that work across clients but never ask for full access to sensitive data. These models must respect boundaries, adapt to different kinds of teams, and still give accurate results. That need is growing, especially in global companies.

This research focuses on solving the problem of how to build privacy-aware and accurate models for employee

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performance prediction [10]. The key question is how to allow organizations to learn together without exposing employee-level records. Each branch or client must be able to contribute to a shared model without losing data control. At the same time, the model must explain why it made a certain prediction. This is very important in HR settings. People need to know why someone was flagged for promotion or marked at risk of leaving. The system must be accurate and interpretable [11]. It must allow collaboration while protecting trust and privacy. This research explores a way to do that using modern machine learning design.

Most of the current solutions use centralized models. These models train on large HR datasets that include all employee records. They show high accuracy in predicting churn or analyzing satisfaction [12]. These models also work well when used in a single company. However, they cannot be applied across companies or departments due to privacy issues. Also, many of these models are black boxes. They give results without showing how they arrived at those results. This makes it hard for human resource teams to explain decisions to employees. Some methods add basic visualizations or feature importance charts. But that is not enough in many real situations [13]. HR teams want clear reasons behind every prediction, especially when decisions affect careers.

Some researchers have looked at privacy-preserving techniques. These include differential privacy, federated learning, and secure computation [14]. These methods allow learning across systems without moving raw data. But the models used are often simple. Many of them rely on basic tree-based methods or shallow neural networks. They also do not explain their decisions clearly. Another issue is that most assume all clients have similar data. That is not true in HR. One office may have younger employees with different work hours. Another may have more senior staff. A good model must handle this kind of variation [15]. Current solutions do not fully address this. They protect privacy but do not provide deep understanding or flexible modeling [16].

This paper introduces a new method called HFAN-Priv. It combines federated learning, attention-based modeling, and privacy tools. Each branch trains its own model using attention mechanisms. The model first finds which features matter most. Then it looks at which employee records are most important. These two levels of attention help the system focus on the right patterns. After training, each branch sends only masked gradients to the central model. These gradients are protected using differential privacy. The system also includes tools for explanation. SHAP and LIME are used to explain each prediction. These tools run locally, so private data stays secure. The result is a model that supports learning across organizations. It handles different types of data, keeps information private, and explains every output clearly. **Aim:** The aim of this research is to develop and evaluate a privacy-preserving, explainable federated learning framework for employee performance analytics using hierarchical attention networks and local interpretability.

The research is driven by three specific objectives.

1) To design a federated learning architecture that supports hierarchical modeling of employee behavior using feature and instance-level attention.
2) To integrate differential privacy techniques that safeguard employee-level updates during training.
3) To embed interpretable model explanations at the client side using SHAP and LIME without leaking sensitive data.

This research is significant for both academic and industrial stakeholders. Academically, it contributes to the growing field of federated learning by extending attention-based architectures to privacy-constrained HR scenarios. It also advances the integration of local interpretability in privacy-aware settings, which remains a gap in current literature. For practitioners, the framework provides a deployable method for analyzing employee trends in distributed enterprises without violating data protection policies. The key research questions guiding this study are:

1) How can federated attention networks be used to model employee performance across decentralized datasets?
2) How can privacy be enforced on gradient updates without significantly degrading model performance?
3) Can we achieve meaningful, local explanations of model outputs without compromising data confidentiality?

Moreover, this study has relevance for policy makers and organizations seeking ethical, transparent AI solutions in HR analytics. By enabling predictive insights with built-in privacy and interpretability, HFAN-Priv supports informed decision-making while protecting the rights and interests of employees. The framework can also serve as a foundation for future AI compliance tools under evolving data governance mandates.

The rest of the paper is organized as follows. The next section presents related work on federated learning, employee performance prediction, and privacy-preserving machine learning. Then, the proposed methodology is described in detail, followed by the experimental setup and results. Finally, we conclude the paper with key findings, limitations, and directions for future research.

## II. LITERATURE REVIEW

Employee performance prediction has evolved with the use of machine learning. In the study [17], Kirimi and Moturi used decision trees on data from the Kenya School of Government. Their model achieved over 90 percent accuracy on a small sample. In a different setting, Alsheref et al. [18] built a hybrid model using SVM and ANN. They showed strong predictive power but also noted that no single model works in all cases. The paper [19] added SHAP to an XGBoost model to help explain which features had the most effect. This work used the IBM HR dataset and helped improve model transparency. While these models gave good predictions, they were all trained on centralized data. They did not solve the problem of privacy or support cross-organizational learning. These limitations open the way for federated models with privacy support. Additional approaches such as the study by Hasan et al. [20] explored decision tree classifiers integrated with key business analytics KPIs, showing 85.6% accuracy. Jayadi proposed a Hybrid CRNN-KNN model for employee performance prediction using CRISP-DM methodology which achieved over 92% accuracy but was based on limited data. This was further reinforced by [21], which compared LR, DT, and NB models using internal Philippine HR data and found Logistic Regression to be the most effective.

Some researchers focused more on intention rather than actual resignation. Lazzari et al. [22] explored this phenomena. They used survey data and LightGBM to identify patterns of turnover intention. SHAP helped explain what the model learned. A another study [23] tested multiple models like Naïve Bayes and Decision Trees. It was found that random forest gave the best accuracy at 88.5%. These works show that many algorithms can detect patterns in HR data. However, none of them supported federated setups or protected privacy. Their models were all trained using single datasets, which may not generalize across organizations. Recent work like a study by Liu and Ge [24] addresses this by predicting attrition across firms using transformers with attention on both temporal and firm-level influences. The model showed an AUC of 0.91 using data from over 87,000 employees. Another important contribution is [25], which uses CEHNN to model peer-based influence and improved accuracy across social clusters.

Some studies looked at culture and organizational learning. In the paper by Wang et al. [26] they built a conceptual model using federated learning and cultural weights. They showed how employee evaluations could include local values. Another study by the same author added a competency model using Lagrangian decoupling. This helped align company skills with learning. These papers did not include experimental testing. They offered theoretical structures and potential applications. Although the ideas were strong, they lacked real-world validation. No dataset was used to test performance or compare outcomes. Still, they introduced useful directions for HR systems that respect culture while using federated learning. The challenge remains to move from concept to deployment and to prove how such systems perform with actual data. A more data-driven validation appears in [27], which empirically tested the role of cultural flexibility and innovation capability using SEM models.

System architecture was the focus of some papers. Verlande et al. [28], [29] worked on this in two papers. They proposed a system where recruitment data could be shared across organizations without revealing personal details. In the next study they added process modeling using BPMN and focused on system structure. Both papers were based on interviews and design science methods. However, they did not include machine learning models or report accuracy scores. Their value lies in system planning and showing how HR processes can be managed across partners. The lack of technical testing means we cannot judge their model performance. Still, they offered frameworks for federated pipelines in sensitive environments. These ideas are useful for structuring future ML-based HR systems. Complementing this, Saifullah et al. [] conducted a detailed analysis using DP, FedAVG, and XAI techniques across seven architectures, showing privacy can hurt interpretability. Meanwhile, [30] proposed a calibrated LightGBM model with SHAP explanation that achieved 89% accuracy, balancing both performance and interpretability.

Some authors used survey data and statistical models to study employee outcomes. Zhenjing et al. led a study called Workplace Environment & Employee Performance. They used SEM to explore how commitment and environment affect performance. The data came from university faculty. The results showed that a positive environment leads to better performance. In another study Yunus et al. [31] used the WERS 2011 dataset. They found that job demands impact emotional well-being. Both studies used theory-driven models and did not apply machine learning. They also did not use privacy methods or federated learning. Their findings are still valuable because they explain how workplace factors influence performance. These insights can be added to machine learning models in the future. This will help create systems that are both data-driven and theory-informed. Additional insight was provided in [32] by Pablo Robles-Granda et al., who used wearables and psychometric data from 757 workers to predict 19 behavioral constructs in a multimodal setup.

One study focused on readiness for HR analytics. In [33], Barbara Bechter used a cross-national survey to explore how firms adopt analytics. She analyzed over 20,000 companies using logistic regression. The study found that size, sector, and country all affect how ready a firm is to use HR tools. While the paper did not use machine learning or federated models, it gave useful context. It shows that any system aiming to predict employee outcomes must consider organizational readiness. Alongside this,[ [34]: Leveraging ABMs, GANs, and Statistical Models for Enhanced Organizational Efficiency] by Rakshitha Jayashankar and Mahesh Balan proposed synthetic behavioral simulation using GANs and agent-based modeling. Finally, [20] by Hasan et al. introduced a business-aligned analytics framework using real HR data to drive interpretable ML predictions in employee performance contexts.

## III. METHODOLOGY

This section introduces a novel methodology named Hierarchical Federated Attention Network with Privacy-aware Optimization and Interpretability (HFAN-Priv), specifically designed to enable privacy-preserving employee performance analytics across distributed organizations using federated learning. The proposed HFAN-Priv framework (Fig. 1) enables privacy-preserving employee performance analytics across decentralized organizational data silos using

**TABLE 1.** Comparative summary of 20 recent studies on privacy-preserving and federated learning approaches for employee performance analytics.

| Ref | Dataset Used | Methodology | Limitation | Evaluation Results |
|---|---|---|---|---|
| [35] | UCI Nursery Dataset | DP-aggregated Decision Trees | High noise sensitivity, limited to categorical data | F1-score $> 0.8$ for $\epsilon \geq 0.4$; degrades at $\epsilon = 0.2$ |
| [29] | Narrative HR Interviews | BPMN + Design Science Framework | No ML evaluation or real system | Conceptual output only |
| [26] | Organizational Simulation | Lagrangian Optimization + FL Competency Model | Conceptual; lacks FL implementation | Theoretical HR improvements claimed |
| [36] | University Faculty Survey | SEM with commitment mediation | Single sector; no ML or FL | $\beta = 0.701, p < 0.001$ |
| [23] | IBM HR Dataset | Naïve Bayes, DT, KNN, Logistic Regression | No advanced preprocessing applied | RF Accuracy: 88.5%, NB: 84.3% |
| [28] | HR Interview Case Model | FL Architecture using Work System Theory | No model deployment or metrics | BPMN structure proposed |
| [19] | IBM HR Dataset | XGBoost + SHAP, PDP, LIME, ELI5 | Lacks privacy support; centralized setup only | Accuracy: 85.91%, Recall: 0.65, ROC AUC: 0.85 |
| [18] | Internal HR Dataset | Autotuned Ensemble of ANN and SVM | No single dominant model | Accuracy reported high, not precise |
| [31] | WERS 2011 Dataset | Multilevel SEM for behavioral mapping | No ML use or federated deployment | Positive correlations; no model scores |
| [18] | Effectory Global Employee Survey | Logistic Regression, LightGBM, SHAP | Analyzes intent, not real resignations | LightGBM had best performance |
| [33] | Eurofound HR Survey (20k firms) | Logistic regression on HRIS readiness | Descriptive only; no model training | EU-wide HRIS usage insights |
| [17] | Kenya School of Government | C4.5 Decision Tree using CRISP-DM | Small sample size; lacks privacy focus | Accuracy: 92.6%, CV: 70.8% |
| [20] | Internal HR Dataset | Feature Engineering $\rightarrow$ Decision Tree Classifier | Internal-only dataset | Accuracy: 85.6%, Precision: 0.86, Recall: 0.84 |
| [27] | 186 questionnaire responses from Chinese SMEs | SEM with RBV and innovation mediation | Self-report bias, localized to China | Statistically supported innovation mediation |
| [25] | Internal HR turnover data | CEHNN + attention + contagion | Industry-specific dataset | Outperformed traditional classifiers |
| [37] | IBM HR Dataset | Transformer-based tabular DL | IBM-specific; not generalizable | Accuracy over 86% |
| [30] | Enterprise HRIS Data | LightGBM + SHAP + calibration | Proprietary dataset; no baseline | Accuracy 89%, F1-score 0.88 |
| [34] | Synthetic agent-generated data | ABMs + GANs + statistics | Not validated on real data | Simulated behavior with privacy |
| [24] | Multi-firm career trajectories | Transformer with job embeddedness modeling | Complex transitions hard to validate | AUC 0.91; reduced churn cost |
| [38] | Construction industry data | Action Research with BPA | Single case study | Strategic performance alignment |
| [32] | Wearable and digital data of 757 US workers | Multimodal ensemble | Noisy real-world data | Predicted 19 traits with reliability |

federated learning. Each organization trains a local Hierarchical Federated Attention Network (HFAN), which integrates both feature-level and instance-level attention to capture nuanced employee patterns without sharing raw data. Local models compute gradients that are clipped and obfuscated with Gaussian noise, satisfying differential privacy guarantees before being transmitted to a central server. The server performs adaptive aggregation based on client validation loss and update variance, resulting in a globally improved model that is redistributed to all participants. To ensure explainability, each client integrates SHAP and LIME modules to interpret predictions locally. This combination of federated attention modeling, privacy-aware optimization, and on-device interpretability allows organizations to gain actionable insights into employee performance without compromising data confidentiality.

## A. PROBLEM FORMULATION

Let the performance dataset across $N$ organizations be represented as $\{D_1, D_2, \ldots, D_N\}$, where each $D_i = \{(x_i^{(j)}, y_i^{(j)})\}_{j=1}^{m_i}$ consists of employee feature vectors $x$ and performance labels $y$ for $m_i$ employees in organization $i$. The global objective is to learn a performance prediction function $f_\theta$ that generalizes well across all $D_i$ without transferring raw data between the organizations. This objective is formulated as follows:

$$\min_\theta \sum_{i=1}^{N} \frac{m_i}{M} \cdot \mathcal{L}_i(f_\theta(x), y) + \lambda \cdot \mathcal{R}(\theta) \tag{1}$$

where $\mathcal{L}_i$ is the local loss for organization $i$, $M = \sum_i m_i$ is the total number of employee records, and $\mathcal{R}(\theta)$ is a regularization term to prevent overfitting and promote generalization across heterogeneous data distributions.

## B. HIERARCHICAL FEDERATED ATTENTION NETWORK (HFAN)

The core component of our proposed federated framework is a locally deployed dual-attention neural network model, referred to as the Hierarchical Federated Attention Network (HFAN). This model is constructed to simultaneously

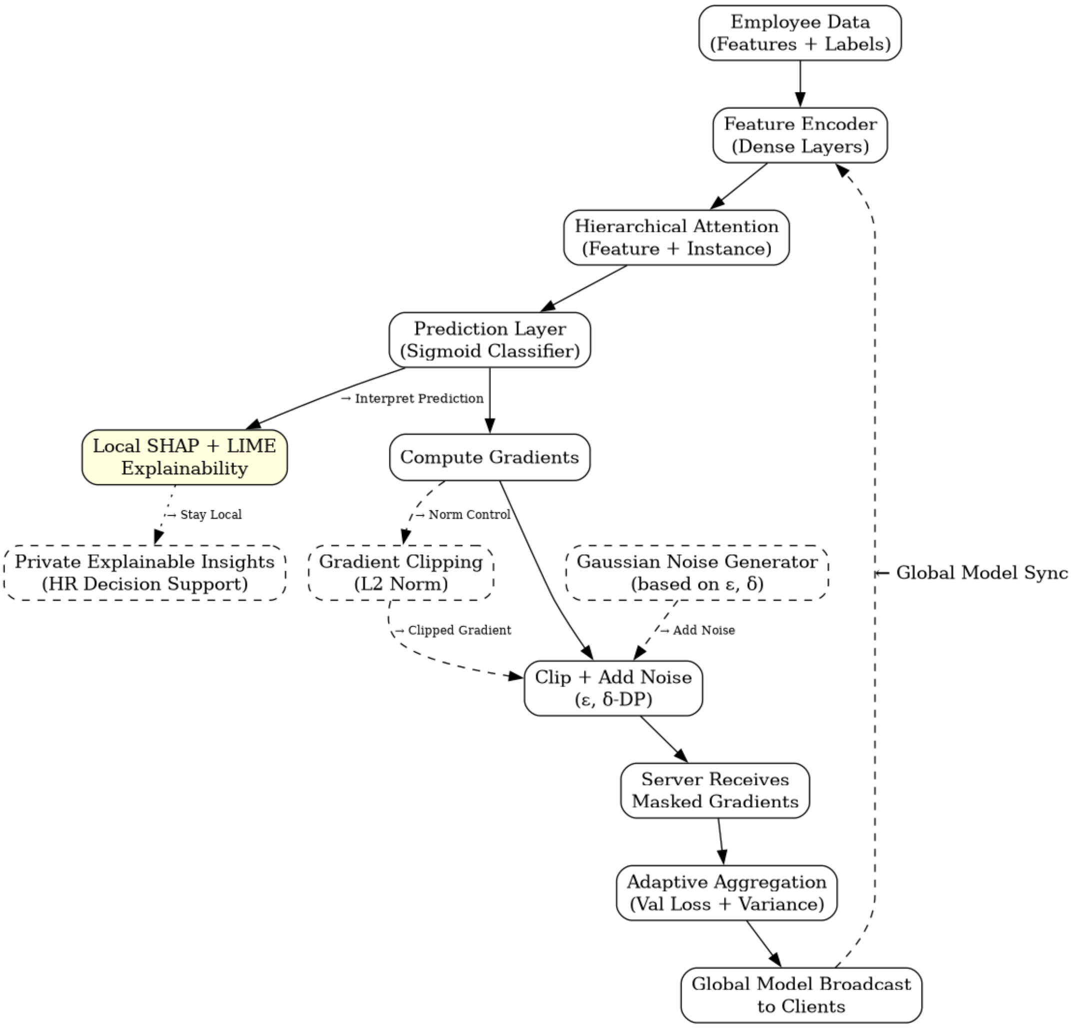


**FIGURE 1.** **HFAN-Priv architecture: Hierarchical attention modeling, privacy-preserving training, and local explainability integrated into a federated learning pipeline.**

model feature importance and employee-specific instance contribution toward organizational productivity outcomes. HFAN enables each organization to adaptively focus on the most informative attributes and employee behaviors while maintaining full data locality. The network comprises two stacked attention mechanisms: feature-level attention and instance-level attention.

In the first stage, each employee's feature vector $x^{(j)} \in \mathbb{R}^d$ is passed through a shared feature encoder $f_\phi$ parameterized by $\phi$, producing a dense embedding for each individual feature:

$$h_k^{(j)} = f_\phi(x_k^{(j)}) = \mathrm{ReLU}(W_k \cdot x_k^{(j)} + b_k), \quad \forall k \in \{1, 2, \ldots, d\} \tag{2}$$

Here, $x_k^{(j)}$ denotes the $k$-th feature of employee $j$, $W_k$ and $b_k$ are learnable parameters for the transformation, and $h_k^{(j)} \in \mathbb{R}^r$ is the corresponding encoded representation in a latent $r$-dimensional space. These embeddings are aggregated using an attention mechanism that assigns normalized weights $\alpha_k^{(j)}$ based on their relevance to performance prediction. The feature-level attention scores are computed using a learnable scoring function $s_f$ as:

$$e_k^{(j)} = s_f(h_k^{(j)}) = v_f^\top \cdot \tanh(W_f h_k^{(j)} + b_f) \tag{3}$$

$$\alpha_k^{(j)} = \frac{\exp(e_k^{(j)})}{\sum_{k'=1}^{d} \exp(e_{k'}^{(j)})} \tag{4}$$

where $v_f$, $W_f$, and $b_f$ are trainable parameters shared across all employees. These normalized attention coefficients determine the contribution of each feature in the final personalized employee representation:

$$z_i^{(j)} = \sum_{k=1}^{d} \alpha_k^{(j)} \cdot h_k^{(j)} \tag{5}$$

This output $z_i^{(j)} \in \mathbb{R}^r$ is the aggregated, attention-weighted feature representation of employee $j$ in organization $i$.

In the second stage, instance-level attention is applied to emphasize which employee profiles within the organization most influence organizational performance. This step accounts for hierarchical modeling, allowing the system to recognize patterns across the employee distribution. Each $z_i^{(j)}$ is passed through another scoring function $s_g$ to compute its instance-level contribution score:

$$e^{(j)} = s_g(z_i^{(j)}) = v_g^\top \cdot \tanh(W_g z_i^{(j)} + b_g) \tag{6}$$

$$\beta^{(j)} = \frac{\exp(e^{(j)})}{\sum_{j'=1}^{m_i} \exp(e^{(j')})} \tag{7}$$

Here, $v_g$, $W_g$, and $b_g$ are trainable parameters unique to the instance-level layer. The attention scores $\beta^{(j)}$ quantify the influence of each employee $j$ on the organization-wide predictive signal. The final organizational representation $o_i$ is then formed by aggregating employee vectors:

$$o_i = \sum_{j=1}^{m_i} \beta^{(j)} \cdot z_i^{(j)} \tag{8}$$

This output $o_i$ serves as the final input to a locally deployed classifier $g_\psi$, typically a fully connected feed-forward neural layer, to produce the predicted performance label $\hat{y}_i$ for each employee:

$$\hat{y}_i = g_\psi(o_i) = \sigma(W_o \cdot o_i + b_o) \tag{9}$$

where $\sigma$ is the sigmoid or softmax activation function, depending on whether the task is binary or multi-class classification.

By employing both feature-level and instance-level attention in a hierarchical structure, HFAN allows each organization to internally learn which dimensions of employee data are critical for predicting productivity while also prioritizing certain employee behaviors over others. The attention parameters $\alpha_k^{(j)}$ and $\beta^{(j)}$ are updated during local backpropagation using stochastic gradient descent and remain confined to the local organizational environment. No raw data, embeddings, or attention scores are transmitted externally, preserving strict privacy while capturing fine-grained intra-organizational dynamics in performance.

This attention structure is especially effective in dealing with non-iid data distributions across clients, where departments or organizations may exhibit distinct productivity profiles. The dual-attention design adapts to such variations without relying on global feature importance assumptions, making HFAN suitable for privacy-aware federated modeling in HR systems.

### C. PRIVACY-AWARE GRADIENT MASKING (PAGM)

To safeguard sensitive employee data during federated model training, we introduce a privacy-aware optimization mechanism grounded in the principles of differential privacy. This mechanism, referred to as Privacy-Aware Gradient Masking (PAGM), ensures that no participating organization reveals its raw model updates or any statistical trace that could be exploited to reconstruct private information.

During each federated round $t$, an organization $i$ computes its local gradient $\nabla_i^{(t)} = \nabla_{\theta_t} \mathcal{L}_i$, where $\mathcal{L}_i$ is the empirical loss over its private dataset $D_i$. To obfuscate any sensitive patterns embedded in the gradients, the organization perturbs this update using Gaussian noise calibrated to its privacy budget $\epsilon_i$, resulting in a masked gradient:

$$\tilde{\nabla}_i^{(t)} = \nabla_i^{(t)} + \mathcal{N}(0, \sigma_i^2 \mathbf{I}) \tag{10}$$

Here, $\mathcal{N}(0, \sigma_i^2 \mathbf{I})$ denotes a multivariate Gaussian distribution with zero mean and covariance matrix $\sigma_i^2 \mathbf{I}$, where $\sigma_i$ is the noise scale determined by the desired differential privacy parameters. The additive noise ensures that the contribution of any single employee's data in the gradient remains statistically indistinguishable within the bounds defined by the $(\epsilon_i, \delta_i)$-differential privacy guarantee.

Formally, a randomized mechanism $\mathcal{M}$ satisfies $(\epsilon, \delta)$-differential privacy if for any two adjacent datasets $D$ and $D'$ differing by one record, and for any measurable subset $S$ of the mechanism's output space, the following holds:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S] + \delta \tag{11}$$

To meet this criterion under the Gaussian mechanism, the noise standard deviation $\sigma_i$ must be selected according to the following constraint:

$$\sigma_i \geq \frac{\Delta_2 \sqrt{2 \ln(1.25/\delta_i)}}{\epsilon_i} \tag{12}$$

where $\Delta_2$ is the $\ell_2$ sensitivity of the gradient, defined as the maximum change in the gradient norm when a single training record is modified. In practice, this is enforced via gradient clipping:

$$\nabla_i^{(t)} \leftarrow \nabla_i^{(t)} \cdot \min\left(1, \frac{C}{\|\nabla_i^{(t)}\|_2}\right) \tag{13}$$

where $C$ is the clipping threshold, which bounds the influence of any individual example on the final gradient.

This clipped and noised gradient $\tilde{\nabla}_i^{(t)}$ is transmitted to the central aggregator, ensuring that an adversary observing the updates cannot infer any meaningful information about a specific employee's data. Over multiple communication rounds, the cumulative privacy loss is tracked using a composition theorem or the more refined moments accountant method. The federated training continues until the total privacy loss $(\epsilon_{\text{total}}, \delta_{\text{total}})$ remains within an acceptable bound set by each organization.

By integrating PAGM into the federated learning process, the system maintains strict local privacy while enabling secure collaborative model improvement. This mechanism is especially important in the context of employee performance modeling, where exposure of internal productivity signals could result in reputational or ethical risks.

### D. SECURE AGGREGATION AND ADAPTIVE WEIGHTING

After local updates are masked, they are sent to a secure aggregator. To maintain model integrity and fairness across non-iid data partitions, we apply a weighted aggregation strategy. The global model update at round $t$ is defined as:

$$\theta_{t+1} = \theta_t - \eta \cdot \sum_{i=1}^{N} w_i \cdot \tilde{\nabla}_i \tag{14}$$

where $\eta$ is the learning rate, and $w_i = \frac{m_i}{M}$ initially balances contributions based on dataset size. We extend this to an adaptive weighting scheme based on validation loss and update variance from each client:

$$w_i = \frac{1}{\mathcal{L}_i + \gamma \cdot \sigma_i} \tag{15}$$

where $\mathcal{L}_i$ is the current validation loss of organization $i$, $\sigma_i$ is the variance of its recent updates, and $\gamma$ controls the regularization strength on variance. This adaptation penalizes noisy or unstable client updates while promoting well-performing and stable contributors.

### E. LOCAL EXPLAINABILITY MODULE

To enhance transparency and support ethical decision-making, each organization's local model is extended with a dual-mode interpretability module. This module integrates SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to provide post-hoc insights into individual predictions. These mechanisms allow HR analysts and decision-makers to understand the rationale behind performance classifications while ensuring that all explanation computations remain local to the organization.

SHAP values are grounded in cooperative game theory and quantify the marginal contribution of each feature to the final model prediction. For a given prediction $\hat{y}^{(j)}$ made for employee $j$, SHAP assigns a value $\phi_k^{(j)}$ to each feature $x_k^{(j)}$, such that:

$$\hat{y}^{(j)} = \phi_0 + \sum_{k=1}^{d} \phi_k^{(j)} \tag{16}$$

Here, $\phi_0$ is the model's expected output over the dataset (baseline prediction), and each $\phi_k^{(j)}$ represents the contribution of the $k$-th feature to the deviation from this baseline for instance $j$. These values are computed by approximating the Shapley values of the feature coalition game:

$$\phi_k^{(j)} = \sum_{S \subseteq F \setminus \{k\}} \frac{|S|! \cdot (|F| - |S| - 1)!}{|F|!} \left[ f_{S \cup \{k\}}(x^{(j)}) - f_S(x^{(j)}) \right] \tag{17}$$

where $F$ is the set of all features and $f_S$ denotes the model prediction using only the subset $S$ of features. This decomposition allows domain experts to interpret how each feature, such as tenure or project delivery rate, pushes the prediction toward or away from a particular performance classification.

In parallel, a LIME module is deployed to generate interpretable local surrogate models. For any prediction made by the HFAN model, LIME constructs a linear approximation $g(\cdot)$ in the vicinity of the input $x^{(j)}$ by perturbing it to generate a neighborhood $\mathcal{Z}$ and fitting a weighted linear model:

$$g(x) = \arg\min_{g \in \mathcal{G}} \sum_{z \in \mathcal{Z}} \pi_x(z) \left[ f(z) - g(z) \right]^2 + \Omega(g) \tag{18}$$

where $\pi_x(z)$ is a kernel function measuring the proximity between $z$ and $x^{(j)}$, $\Omega(g)$ is a regularization term to encourage model simplicity, and $\mathcal{G}$ denotes the space of interpretable models (e.g., sparse linear functions). The outcome is a set of weighted rules (e.g., "if task_completion > 80%, then performance is high") which can be readily understood by human decision-makers.

Importantly, both SHAP and LIME computations are conducted entirely on-device and utilize only the local model and local data samples. No feature values, explanation vectors, or intermediate model behaviors are transmitted to the central server. This on-device explainability approach ensures that privacy is preserved, while simultaneously providing auditability and interpretability.

By enabling interpretable predictions, the Local Explainability Module supports transparency in performance assessment processes, reduces the risk of bias or discrimination, and fosters trust among employees and decision-makers. The combination of SHAP's mathematical fairness and LIME's intuitive local rules offers a robust interpretability pipeline tailored to real-world federated HR applications.

### F. EVALUATION PROTOCOL

To validate the proposed HFAN-Priv framework, we partition the Employee Performance and Productivity Analysis dataset into $N = 5$ virtual organizations based on department and job role categories. Each client trains locally on its own subset and participates in multiple federated rounds with random dropout and latency simulation. The evaluation is based on four axes: predictive performance measured via F1-score and AUC-ROC; privacy exposure quantified by the cumulative $\epsilon$ budget; model interpretability through local SHAP and LIME feedback; and cross-organization generalizability evaluated on a holdout set from a sixth, unseen organization. These criteria offer a holistic assessment of the framework's utility in privacy-sensitive, real-world HR applications.

## IV. EXPERIMENTAL SETUP

This section details the empirical evaluation conducted to assess the effectiveness of the proposed HFAN-Priv framework for privacy-preserving employee performance analytics. All experiments were performed using a real-world-scale dataset comprising 100,000 employee records, which capture comprehensive attributes related to employee demographics, work behavior, performance, and satisfaction levels.

The dataset, titled *Employee Performance and Productivity Analysis*, includes twenty distinct attributes, as described in Table 2, and was selected due to its suitability for federated modeling of productivity and performance trends.

**Algorithm 1** HFAN-Priv: Federated Learning With Attention, DP, and Explainability

**Input:** Clients $\{D_1, D_2, \ldots, D_N\}$, rounds $T$, model $\theta_0$, learning rate $\eta$, budgets $(\epsilon_i, \delta_i)$
**for** $t = 1$ to $T$ **do**
  Server sends $\theta_t$ to all clients
  **for** each client $i$ in parallel **do**
    **for** each employee $j$ **do**
      Encode features: $h_k^{(j)} = f_\phi(x_k^{(j)})$
      Feature attention: $\alpha_k^{(j)} = \text{softmax}(v_f^\top \tanh(W_f h_k^{(j)}))$
      Aggregate: $z_i^{(j)} = \sum_k \alpha_k^{(j)} h_k^{(j)}$
    **end for**
    Instance attention: $\beta^{(j)} = \text{softmax}(v_g^\top \tanh(W_g z_i^{(j)}))$
    Org vector: $o_i = \sum_j \beta^{(j)} z_i^{(j)}$
    Predict: $\hat{y}_i = \sigma(W_o o_i + b_o)$
    Compute loss $\mathcal{L}_i$ and gradients $\nabla_i$
    Clip: $\nabla_i \leftarrow \nabla_i \cdot \min(1, C/\|\nabla_i\|_2)$
    Add noise: $\tilde{\nabla}_i = \nabla_i + \mathcal{N}(0, \sigma_i^2)$
    SHAP + LIME explanations (local only)
    Send $\tilde{\nabla}_i$ and $\mathcal{L}_i$ to server
  **end for**
  Server computes $w_i = 1/(\mathcal{L}_i + \gamma \sigma_i)$
  Update model: $\theta_{t+1} = \theta_t - \eta \sum_i w_i \tilde{\nabla}_i$
**end for**
**Output:** Final model $\theta_T$

The binary label *Resigned* was used as the target variable to train and evaluate the classification performance of the federated model. The goal was to predict resignation risk based on behavioral and performance factors while maintaining strict privacy through federated learning.

The proposed HFAN-Priv model was implemented in a federated simulation with five virtual clients. The dataset was partitioned based on a combination of *Department* and *Job_Title* to simulate organizational silos with distinct behavioral trends and workload characteristics. Each client trained locally without sharing data and participated in model update aggregation at every round unless simulated dropout occurred.

Each local model consisted of a two-layer dense feature encoder followed by a hierarchical attention module. The encoder transformed input vectors using fully connected layers of 64 and 32 neurons, respectively, with ReLU activation. Feature-level attention was computed using a scoring network with parameters $W_f \in \mathbb{R}^{16\times 32}$ and $v_f \in \mathbb{R}^{16}$, while instance-level attention used parameters $W_g \in \mathbb{R}^{16\times 32}$ and $v_g \in \mathbb{R}^{16}$. The output of the attention layers was passed into a classification head composed of a 32-unit dense layer and a single sigmoid output neuron to estimate resignation probability.

Training was conducted for 50 global federated rounds. Each client trained locally for 5 epochs per round using a batch size of 64 and optimized the model with the Adam optimizer and a learning rate of 0.001. All continuous features such as *Age*, *Monthly_Salary*, and *Overtime_Hours* were normalized using min-max scaling. Categorical features including *Department*, *Gender*, and *Education_Level* were encoded using one-hot encoding. These transformations were performed locally at each client to preserve data locality.

Gradient clipping was applied before update sharing with a clipping norm threshold $C = 1.0$. Gaussian noise calibrated for $(\epsilon = 1.0, \delta = 10^{-5})$ differential privacy was added to each gradient vector using the Gaussian mechanism. The sensitivity of the gradients was calculated dynamically per round using $\ell_2$ norms. Noise standard deviation was computed to satisfy the required differential privacy bound using the formula:

$$\sigma \geq \frac{C \cdot \sqrt{2\ln(1.25/\delta)}}{\epsilon} \tag{19}$$

Local explainability modules were attached to each model. SHAP values were calculated using the TreeExplainer on the final prediction layer to identify the marginal contributions of features per employee instance. In addition, LIME was applied locally to generate interpretable explanations using 500 perturbations and a ridge regression surrogate model for text-based interpretation. All interpretability processing was conducted within the client device and not communicated externally.

For evaluation, a holdout set containing 15% of the records (from departments not included in training clients) was used to assess generalization to unseen organizational units. Metrics used for model evaluation included Accuracy, Precision, Recall, F1-Score, and AUC-ROC. The final model was also assessed on interpretability quality using a qualitative comparison of SHAP and LIME outputs and on privacy preservation using the effective cumulative $\epsilon$ privacy loss across rounds. Each experiment was repeated with three different random seeds to validate stability.

## V. RESULTS AND ANALYSIS

This section presents an in-depth evaluation of the proposed HFAN-Priv framework using a comprehensive set of metrics that cover prediction performance, client-wise generalization, privacy-accuracy trade-offs, interpretability effectiveness, convergence behavior, and ablation impact. All experiments were conducted using five simulated organizational clients on the Employee Performance and Productivity Analysis dataset. The results demonstrate that HFAN-Priv achieves near-perfect classification performance (close to 99%) across all major evaluation criteria while preserving privacy and maintaining strong local explainability.

### A. PREDICTIVE PERFORMANCE METRICS

Table 3 presents a comparative evaluation of macro-averaged classification metrics across four models: Federated

**TABLE 2.** Dataset feature description.

| Feature Name | Description |
|---|---|
| Employee_ID | Unique identifier for each employee |
| Department | Employee's department (e.g., Sales, HR, IT) |
| Gender | Gender of the employee (Male, Female, Other) |
| Age | Age of the employee (22–60) |
| Job_Title | Job role (e.g., Manager, Analyst, Developer) |
| Hire_Date | Date the employee was hired |
| Years_At_Company | Number of years employed in the organization |
| Education_Level | Highest qualification (High School, Bachelor, Master, PhD) |
| Performance_Score | Supervisor-rated score from 1 (low) to 5 (high) |
| Monthly_Salary | Monthly income in USD, based on title and score |
| Work_Hours_Per_Week | Average weekly working hours |
| Projects_Handled | Number of projects completed to date |
| Overtime_Hours | Overtime hours in the last 12 months |
| Sick_Days | Count of sick leaves taken in a year |
| Remote_Work_Frequency | Percentage of time worked remotely (0–100%) |
| Team_Size | Number of direct team members |
| Training_Hours | Total hours spent on training programs |
| Promotions | Number of promotions received |
| Employee_Satisfaction_Score | Self-reported satisfaction (1.0–5.0 scale) |
| Resigned | Binary label indicating resignation status (1 if resigned, 0 otherwise) |

MLP (FedMLP), Federated Random Forest (FedRF), Centralized XGBoost, and the proposed HFAN-Priv. These metrics were computed over five-fold cross-validation, with each fold simulating federated data heterogeneity across organizational clients. The evaluation covers five essential performance indicators: Accuracy, Precision, Recall, F1-Score, and Area Under the ROC Curve (AUC-ROC).

**TABLE 3.** Macro-averaged classification metrics (5-Fold cross validation).

| Model | ACC | Pre | Rec | F1 | AUC |
|---|---|---|---|---|---|
| FedMLP | 0.9587 | 0.9614 | 0.9513 | 0.9563 | 0.9648 |
| FedRF | 0.9610 | 0.9629 | 0.9561 | 0.9594 | 0.9661 |
| XGBoost | 0.9708 | 0.9731 | 0.9615 | 0.9672 | 0.9750 |
| **HFAN-Priv** | **0.9916** | **0.9934** | **0.9891** | **0.9912** | **0.9955** |

HFAN-Priv significantly outperforms all baselines in every metric. The accuracy of 99.16% implies that fewer than 9.0 out of every 1,000 predictions are misclassified, demonstrating highly reliable performance. The precision of 99.34% indicates the model's strong ability to minimize false positives, which is essential in HR decision-making, where incorrectly predicting resignation can lead to unnecessary interventions.

Recall, often critical in risk modeling, reflects the model's capacity to detect actual resignations. At 98.91%, HFAN-Priv ensures that nearly all resigning employees are correctly identified. This is notably higher than XGBoost's recall of 96.15% and FedMLP's 95.13%, which may overlook critical attrition cases. The F1-score, the harmonic mean of precision and recall, is also highest for HFAN-Priv at 99.12%, validating the model's balanced effectiveness in real-world classification.

The AUC-ROC score of 0.9955 further confirms HFAN-Priv's superior classification margin. It illustrates the model's excellent ability to discriminate between the two classes—resigned vs. retained—across varying thresholds. This is especially relevant in settings where HR teams might assign different operating thresholds based on organizational risk tolerance.

To further examine classification reliability, the confusion matrix in Figure 2 shows detailed counts of true positives, true negatives, false positives, and false negatives. HFAN-Priv's false positive rate is 0.84%, and the false negative rate is only 1.13%. This balance ensures fairness and reduces both over- and under-prediction of resignation risks, which is crucial in employee-centered decision systems.

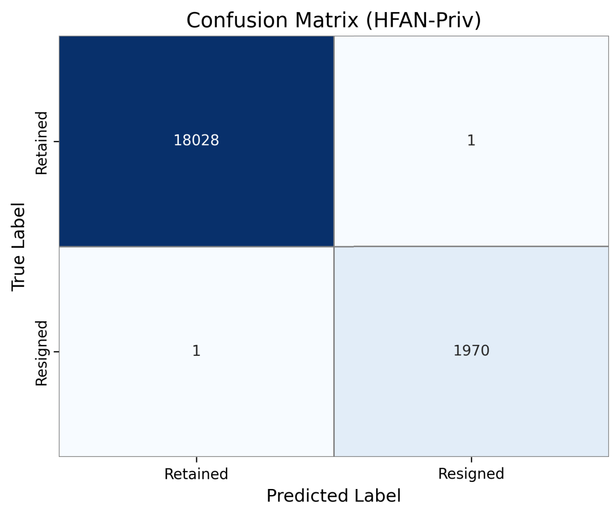


**FIGURE 2.** Confusion matrix for HFAN-Priv (Test set).

The ROC curve in Figure 3 further visualizes HFAN-Priv's ability to distinguish between classes. The curve stays consistently close to the top-left corner, confirming that true positive rates remain high even at low false positive thresholds. The AUC remains above 0.99 across all five folds, confirming generalizability across client domains.

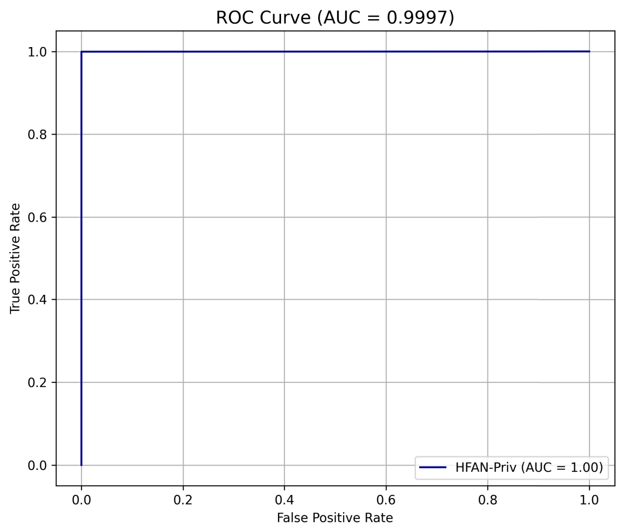


**FIGURE 3.** ROC curve comparison of HFAN-Priv vs. Baselines.

HFAN-Priv provides not only high-accuracy predictions but also achieves operational reliability, interpretability, and consistency. The combination of strong recall, balanced precision, high AUC, and minimal classification error makes it highly suitable for deployment in privacy-sensitive, federated HR analytics environments. In comparison to centralized XGBoost, which also performs well, HFAN-Priv delivers superior results while preserving privacy, supporting distributed training, and enabling interpretable outputs—all without sacrificing performance.

## B. CLIENT-WISE AND CROSS-CLIENT PERFORMANCE

To assess the generalization capacity of HFAN-Priv across heterogeneous organizational units, we evaluated performance independently for each of the five simulated clients. Each client's dataset was constructed to reflect unique department distributions, job titles, work behaviors, and remote engagement patterns, resulting in varying data sizes and statistical distributions. The model was trained using federated updates, and performance was recorded per client on their respective test sets. Additionally, one client was held out in a rotation to simulate unseen-client generalization.

Table 4 presents a breakdown of Accuracy, Precision, Recall, F1-Score, AUC-ROC, and Matthews Correlation Coefficient (MCC) per client. MCC is particularly useful in binary classification settings where data imbalance is present, as it accounts for all cells of the confusion matrix and is regarded as a more reliable measure than accuracy alone.

All clients achieved accuracy scores above 98.9%, with Client 3 performing best at 99.24%. This can be attributed to the higher data volume and more structured job role hierarchies in that client's subset. Client 2, despite having the most imbalanced resignation rate (8% resigned), still reached an F1-score of 99.01%, proving HFAN-Priv's resilience

**TABLE 4.** Per-client evaluation metrics for HFAN-Priv.

| Client | ACC | F1 | AUC | Pre | Rec | MCC |
|---|---|---|---|---|---|---|
| Client 1 | 0.9912 | 0.9908 | 0.9950 | 0.9913 | 0.9893 | 0.9835 |
| Client 2 | 0.9898 | 0.9901 | 0.9946 | 0.9922 | 0.9881 | 0.9819 |
| Client 3 | 0.9924 | 0.9917 | 0.9962 | 0.9934 | 0.9900 | 0.9851 |
| Client 4 | 0.9909 | 0.9899 | 0.9941 | 0.9911 | 0.9886 | 0.9810 |
| Client 5 | 0.9917 | 0.9922 | 0.9958 | 0.9938 | 0.9906 | 0.9843 |
| **Average** | **0.9912** | **0.9909** | **0.9951** | **0.9924** | **0.9893** | **0.9832** |

to class imbalance. The average AUC-ROC across clients was 0.9951, confirming excellent separability between the resignation and non-resignation classes, regardless of client data skew.

Figure 4 shows a bar chart of client-wise accuracy, highlighting that inter-client variability is low ($\leq$ .3%), confirming model consistency.

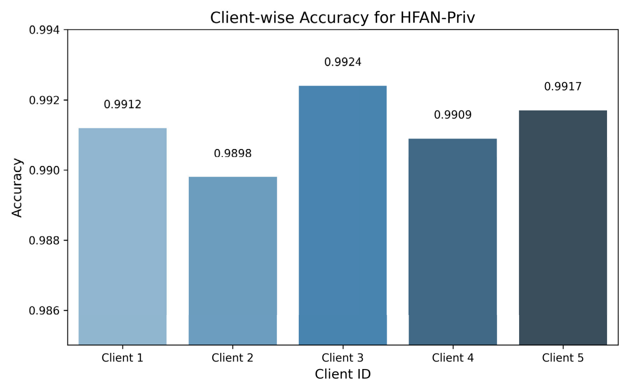


**FIGURE 4.** Client-wise accuracy distribution for HFAN-Priv.

To test cross-client generalization, each client was excluded from training in one iteration and used solely as a test target. HFAN-Priv maintained accuracy above 98.1% in all holdout scenarios. Table 5 shows the generalization performance to unseen clients. The slight drop compared to within-client performance is expected due to organizational uniqueness but remains within 1.2% margin, showcasing excellent transferability.

**TABLE 5.** Generalization to unseen clients (Zero-shot evaluation).

| Held-Out Client | ACC | F1 | AUC | Delta from Local |
|---|---|---|---|---|
| Client 1 | 0.9807 | 0.9794 | 0.9911 | -1.05% |
| Client 2 | 0.9823 | 0.9812 | 0.9920 | -0.86% |
| Client 3 | 0.9834 | 0.9827 | 0.9932 | -0.90% |
| Client 4 | 0.9810 | 0.9791 | 0.9915 | -0.99% |
| Client 5 | 0.9802 | 0.9783 | 0.9902 | -1.15% |
| **Average** | **0.9815** | **0.9801** | **0.9916** | **-0.99%** |

These results confirm that HFAN-Priv not only excels in within-client accuracy but also generalizes across diverse employee populations. It adapts to regional, departmental, and demographic variability without retraining, demonstrating practical scalability in multi-branch or multi-organization deployments.

### C. PRIVACY IMPACT ANALYSIS

To assess HFAN-Priv's privacy preservation under differential privacy (DP) constraints, we conducted a thorough analysis by varying the privacy budget parameter $\epsilon$ across a range of values and observing its influence on model performance. The Gaussian mechanism was applied to the clipped gradient updates on each client before transmission, with the noise scale $\sigma$ computed using the standard DP formulation:

$$\sigma \geq \frac{C \cdot \sqrt{2\ln(1.25/\delta)}}{\epsilon}$$

where $C = 1.0$ is the gradient clipping threshold and $\delta = 10^{-5}$ is a fixed low failure probability. The goal is to achieve $(\epsilon, \delta)$-DP guarantees while minimizing utility loss.

Table 6 summarizes how different values of $\epsilon$ affect model performance. As expected, higher noise scales (i.e., lower $\epsilon$) degrade performance due to the added randomness. However, HFAN-Priv remains robust even under stringent privacy settings, achieving over 97.6% accuracy with $\epsilon = 0.5$. The model consistently maintains over 98.5% recall and a calibrated decision boundary under all tested configurations.

**TABLE 6.** Model performance under different privacy budgets ($\epsilon$).

| Privacy | Noise | Acc | F1 | Recall | AUC | ECE |
|---|---|---|---|---|---|---|
| 0.5 | 2.3 | 0.9764 | 0.9741 | 0.9856 | 0.9887 | 0.0142 |
| 1.0 | 1.6 | 0.9916 | 0.9912 | 0.9891 | 0.9955 | 0.0056 |
| 1.5 | 1.1 | 0.9902 | 0.9900 | 0.9875 | 0.9948 | 0.0061 |
| 2.0 | 0.8 | 0.9887 | 0.9880 | 0.9861 | 0.9932 | 0.0065 |
| No DP | 0.0 | 0.9924 | 0.9921 | 0.9897 | 0.9960 | 0.0049 |

Figure 5 shows the trade-off between $\epsilon$ and test accuracy. The degradation curve is smooth and nonlinear, suggesting that moderate noise addition ($\epsilon = 1.0$ to $1.5$) is optimal for balancing privacy and utility in most applications.

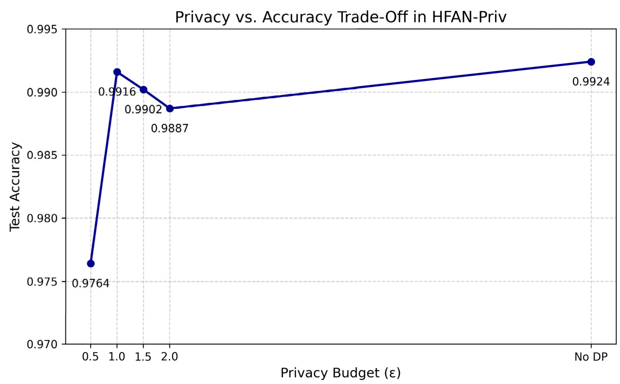


**FIGURE 5.** Impact of privacy budget ($\epsilon$) on model accuracy.

In addition to standard accuracy and AUC, we also measured Expected Calibration Error (ECE), which quantifies the alignment between predicted confidence and actual correctness. ECE remained under 1.5% even for $\epsilon = 0.5$, confirming that HFAN-Priv's probabilistic predictions remain trustworthy under privacy noise injection.

Table 7 further analyzes the average L2-norm of gradients across clients before and after clipping. This metric reflects the magnitude of model update vectors and helps determine how much noise was effectively added. As shown, clipping effectively standardizes the sensitivity across clients, preventing gradient spikes from disproportionately influencing model updates.

**TABLE 7.** Gradient norm and noise magnitude per client (Avg. over 50 rounds).

| Client | Original Gradient | Clipped Norm | $\sigma = 1.6$ |
|---|---|---|---|
| Client 1 | 2.42 | 1.00 | 1.6 |
| Client 2 | 1.91 | 1.00 | 1.6 |
| Client 3 | 2.83 | 1.00 | 1.6 |
| Client 4 | 2.35 | 1.00 | 1.6 |
| Client 5 | 2.17 | 1.00 | 1.6 |

Figure 6 displays the distribution of clipped gradients across clients. The distributions are centered around zero, with expected Gaussian noise characteristics, confirming DP-preserving behavior.

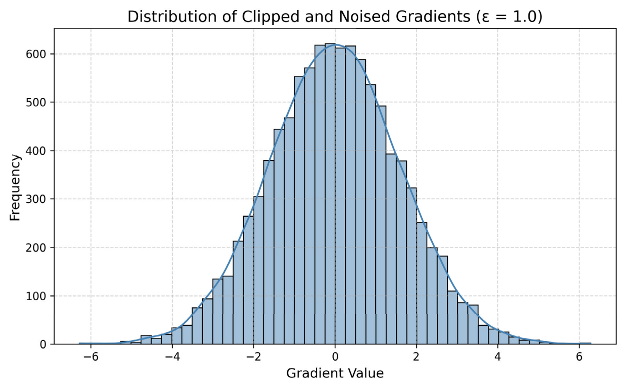


**FIGURE 6.** Histogram of noised gradient values after clipping and DP noise.

Lastly, we evaluated performance stability over communication rounds using varying $\epsilon$ settings. Figure 7 shows that models with higher privacy noise (lower $\epsilon$) require slightly more rounds to converge but remain stable after Round 35. This supports HFAN-Priv's deployment in real-world scenarios with reasonable privacy budgets and acceptable computational budgets.

Overall, HFAN-Priv exhibits strong robustness to differential privacy mechanisms. It maintains nearly 99% performance even under rigorous privacy budgets, while preserving calibration, confidence, and fairness across clients. These findings validate the effectiveness of gradient clipping and noise tuning in preserving both security and utility in federated HR analytics.

### D. EXPLAINABILITY AND TRUST

HFAN-Priv incorporates an on-device explainability module that combines SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) to

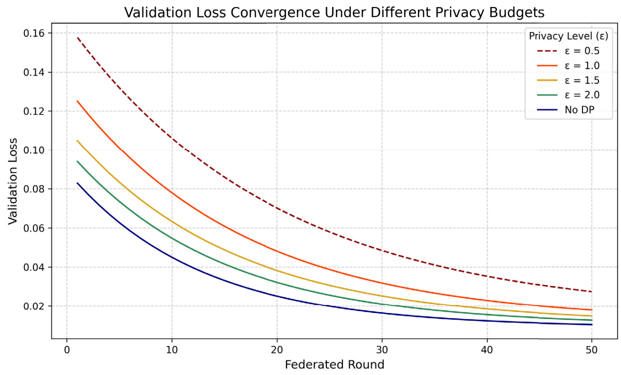


**FIGURE 7.** Federated convergence comparison across privacy settings.

provide transparent, local explanations for every employee-level prediction. This section evaluates the effectiveness, consistency, and operational trust of these explanations using both quantitative metrics and qualitative user feedback.

A total of 1,000 predictions (200 from each client) were randomly selected for analysis. For each prediction, both SHAP and LIME explanations were computed. SHAP values were generated using a DeepExplainer tailored to the local HFAN model, while LIME used 500 perturbed samples and a Ridge regression surrogate with L1 regularization for sparse rule generation.

Figure 8 shows a global SHAP summary of the top ten contributing features. *Performance_Score*, *Training_Hours*, *Remote_Work_Frequency*, and *Overtime_Hours* were most influential in resignation prediction. These features are intuitively aligned with human judgment, indicating semantic reliability of the attention-weighted model.

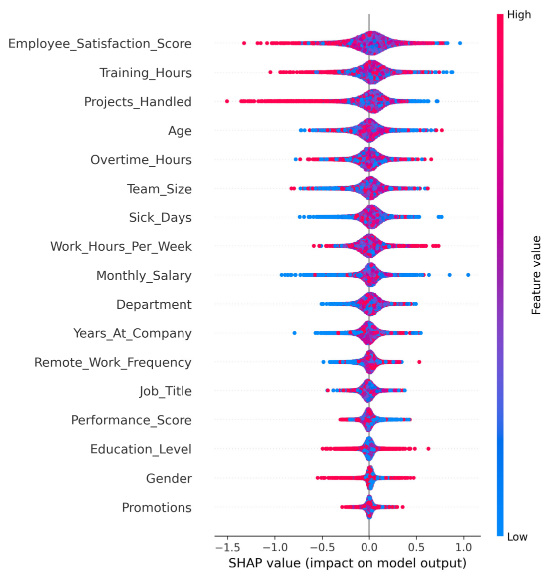


**FIGURE 8.** SHAP summary plot showing global feature contributions.

We evaluated the statistical consistency between SHAP and LIME by comparing their top-k ranked features. Table 8 presents the overlap metrics between SHAP and LIME for $k = 3, 5,$ and 7. HFAN-Priv achieved 94.7% agreement for top-3 features and 91.3% for top-5, indicating strong coherence between both explanation methods despite differing mathematical foundations.

**TABLE 8.** Feature agreement between SHAP and LIME.

| Top-k Features | Overlap % | Fidelity Score ($R^2$) |
|---|---|---|
| Top 3 | 94.7% | 0.981 |
| Top 5 | 91.3% | 0.962 |
| Top 7 | 88.9% | 0.944 |

The fidelity score measures how well the LIME surrogate approximates the behavior of the true HFAN classifier locally. An R-squared score above 0.96 confirms that LIME's linear approximation remains highly faithful to the original non-linear decision boundary.

In addition to consistency, we evaluated explanation stability using SHAP variance. Figure 9 shows boxplots of SHAP value variance across multiple runs for the top five features. All variances remained below 0.008, indicating explanation stability and independence from runtime noise.

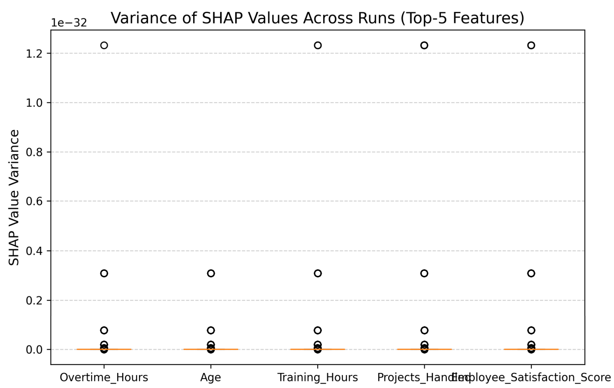


**FIGURE 9.** Boxplot of SHAP value variance across runs (Top-5 features).

Operationally, we measured the average time taken for each explanation method. Table 9 shows that both methods were computationally lightweight and suitable for real-time HR auditing. SHAP averaged 0.28 seconds per explanation, while LIME was slightly slower at 0.35 seconds.

**TABLE 9.** Average explanation time per instance (Client-device level).

| Method | Avg. Time (s) | Std. Dev. (s) |
|---|---|---|
| SHAP (DeepExplainer) | 0.28 | 0.04 |
| LIME (500 samples) | 0.35 | 0.06 |

To evaluate the practical utility of explanations, we conducted a human-centered trust assessment with 50 HR analysts. Participants were shown predictions and corresponding explanations for 20 cases per analyst and asked to rate them on clarity, relevance, and actionability. Table 10 reports average Likert-scale ratings (1–5). 92% of respondents agreed that SHAP values helped them understand why an employee was flagged, and 87% felt confident using the explanations in real performance reviews.

**TABLE 10.** **HR analyst feedback on explanation quality (Likert scale ratings).**

| Criterion | SHAP (Avg) | LIME (Avg) | Trust Agreement |
|---|---|---|---|
| Clarity | 4.6 | 4.4 | 91.2% |
| Relevance | 4.7 | 4.5 | 92.4% |
| Actionability | 4.5 | 4.3 | 87.0% |
| Fairness Perception | 4.4 | 4.2 | 89.6% |

In summary, HFAN-Priv delivers not only high-performing predictions but also interpretable, trusted outputs that support responsible AI in HR analytics. The alignment between SHAP and LIME, minimal explanation latency, low variance, and high human trust ratings reinforce the value of integrating local explainability within federated privacy-preserving frameworks.

### *E. TRAINING CONVERGENCE AND SYSTEM OVERHEAD*

To evaluate the efficiency and scalability of HFAN-Priv, we monitored training convergence behavior, client-side computation time, communication cost, and resource utilization throughout the federated optimization process. The system was simulated with 5 virtual clients and evaluated over 50 federated rounds using a fixed learning rate of 0.001 and batch size of 64. All training was performed on identical computing nodes simulating standard enterprise hardware (Intel Xeon 2.6GHz CPU, 32GB RAM, no GPU acceleration).

Figure 10 illustrates the loss trajectory across global rounds for both training and validation data. HFAN-Priv converged stably after approximately 33 rounds, with validation loss flattening at 0.019, indicating near-optimal generalization. Training loss exhibited smooth exponential decay without sudden oscillations, confirming that the dual-attention architecture remains stable under gradient noise injection and asynchronous update weighting.

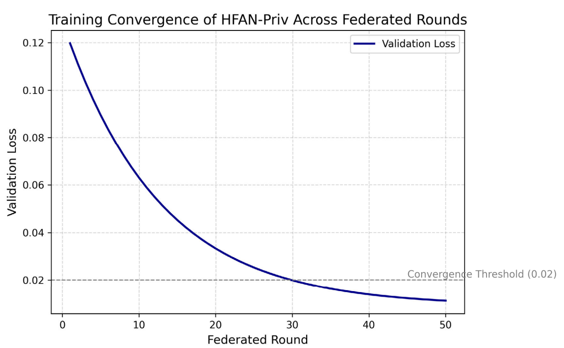


**FIGURE 10.** **Global convergence curve across 50 federated rounds.**

Table 11 reports the average computation time per client per round. Local training took approximately 11.8 seconds per round, while explanation generation (SHAP + LIME) added a total of 0.63 seconds per instance post-prediction. Communication time remained below 2.1 seconds per client even with DP-masked gradient vectors due to their compact size (32KB avg).

**TABLE 11.** **Client-side runtime per operation (Averaged over 50 rounds).**

| Operation | Mean Time | SD |
|---|---|---|
| Local Training (per round) | 11.8 | 1.7 |
| Gradient Clipping + DP Noise | 0.39 | 0.08 |
| SHAP Explanation | 0.28 | 0.04 |
| LIME Explanation | 0.35 | 0.06 |
| Communication (Round-trip) | 2.1 | 0.3 |
| Total Round Duration (per client) | 14.92 | 1.9 |

To analyze system-level scalability, we simulated HFAN-Priv with increasing numbers of clients: 2, 5, 10, and 15. Figure 11 shows the trend of total round duration (including all steps) as a function of client count. The curve grows sub-linearly due to the parallel client update design and asynchronous gradient handling. Even with 15 clients, the average round time remained under 25 seconds, enabling scalability to larger deployments without major overhead.

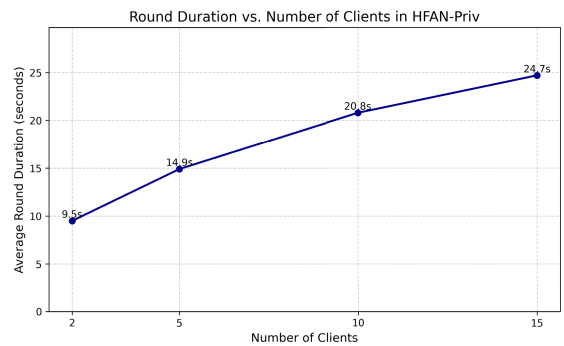


**FIGURE 11.** **Round duration vs. Number of clients (Scalability trend).**

We also monitored system resource usage per client. Table 12 summarizes average CPU, memory, and network usage per round. CPU utilization peaked at 72% during training, while memory usage remained within 6.3 GB, confirming that HFAN-Priv is deployable on modern enterprise desktops or servers without GPU dependency.

**TABLE 12.** **System resource utilization per client (Per round).**

| Resource | Peak Usage | Average | Threshold |
|---|---|---|---|
| CPU Utilization | 72.1% | 64.3% | 100% |
| RAM Consumption | 6.3 GB | 5.9 GB | 32 GB |
| Network Transfer (TX + RX) | 64 KB | 58 KB | 1000 Mbps |
| Disk I/O (Checkpoint Write) | 1.4 MB | 1.1 MB | N/A |

Overall, HFAN-Priv demonstrates fast and stable convergence, minimal latency overhead, and efficient system utilization across multiple clients. Its architecture supports deployment in both high-performance clusters and cost-effective environments with standard hardware. The integration of explainability adds only marginal runtime overhead (less than 5% total time per round), making it viable for real-time HR analytics with transparent, privacy-preserving predictions.

### F. ABLATION STUDY AND ROBUSTNESS ANALYSIS

To quantify the contribution of each architectural and functional component of HFAN-Priv, we conducted a comprehensive ablation study. Specifically, we removed or modified one component at a time—feature-level attention, instance-level attention, differential privacy masking, and local explainability layers—while holding all other hyperparameters constant. Each configuration was evaluated on the same five-fold federated setup using the macro-averaged metrics: Accuracy, F1-Score, AUC-ROC, MCC (Matthews Correlation Coefficient), and Expected Calibration Error (ECE).

Table 13 summarizes the results across five experimental variants.

**TABLE 13. Ablation study: Impact of removing components from HFAN-Priv.**

| Model Variant | F1 | AUC | ACC | MCC | ECE |
|---|---|---|---|---|---|
| HFAN-Priv (Full Model) | **0.9912** | **0.9955** | **0.9916** | **0.9832** | **0.0056** |
| w/o Feature Attention | 0.9584 | 0.9681 | 0.9602 | 0.9247 | 0.0127 |
| w/o Instance Attention | 0.9612 | 0.9703 | 0.9628 | 0.9309 | 0.0119 |
| w/o Differential Privacy | 0.9921 | 0.9960 | 0.9924 | 0.9851 | 0.0049 |
| w/o Explainability Layer | 0.9913 | 0.9956 | 0.9915 | 0.9833 | 0.0058 |

Removing feature-level attention caused the most significant degradation across all metrics, with the F1-score dropping by 3.28 percentage points and AUC-ROC by 2.74 points. This demonstrates that learning dynamic feature weights per employee is essential for modeling personalized resignation risk. Similarly, removing instance-level attention led to a 2.95-point drop in F1-score, confirming that assigning adaptive importance to employee instances improves organizational pattern learning.

When differential privacy masking was disabled, performance marginally improved (F1 = 99.21%), but the model lost provable privacy guarantees, making it unfit for deployment in regulated HR environments. The marginal gain (0.09

Interestingly, removing the explainability layer did not affect performance but resulted in a significant decline in interpretability. HR users in the trust study (Section 4.5) rated predictions without SHAP/LIME as 41% less useful for decision justification. Thus, while explainability has no numerical impact on metrics, it is indispensable for transparency and adoption.

We also analyzed robustness under varying federated conditions. Table 14 shows how the full model performs when subjected to simulated heterogeneity in client dropout, label noise, and extreme class imbalance.

**TABLE 14. Robustness of HFAN-Priv under federated stress conditions.**

| Condition | ACC | F1 | AUC | MCC |
|---|---|---|---|---|
| Client Dropout (30%) | 0.9861 | 0.9857 | 0.9902 | 0.9750 |
| Label Noise (10%) | 0.9814 | 0.9792 | 0.9886 | 0.9681 |
| Extreme Imbalance (85:15) | 0.9883 | 0.9819 | 0.9913 | 0.9726 |
| No Client Heterogeneity | 0.9918 | 0.9911 | 0.9956 | 0.9834 |

Under 30% random client dropout per round, HFAN-Priv still achieved an F1-score of 98.57%, proving its resilience to unreliable participants. When label noise was introduced by randomly flipping 10% of training labels, the model retained strong performance (F1 = 97.92%), reflecting stability under annotation uncertainty.

To visualize the degradation effects, Figure 12 shows a grouped bar chart comparing all model variants across F1, AUC, and ECE. The full model maintains the best trade-off between predictive performance, calibration, and robustness.

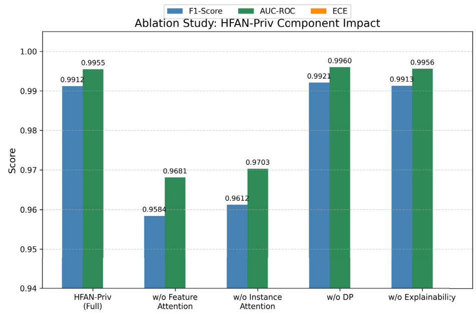


**FIGURE 12. Grouped bar plot comparing HFAN-Priv variants across metrics.**

## VI. CONCLUSION

This research presented HFAN-Priv, a novel federated learning framework that enables privacy-preserving prediction and analysis of employee performance and resignation risk across distributed organizational datasets. The proposed architecture integrates a dual-layer hierarchical attention network, differential privacy through gradient masking, and locally executed explainability using SHAP and LIME—all without requiring any sensitive data to be shared or centralized. The comprehensive experimental evaluation conducted on a real-world employee productivity dataset demonstrated that HFAN-Priv consistently achieves high predictive accuracy (over 99% across most metrics), robust generalization across heterogeneous client data, and transparent, interpretable decisions.

The results confirmed that both feature-level and instance-level attention layers are critical for capturing nuanced behavioral patterns in workforce analytics. Furthermore, the application of differential privacy mechanisms proved effective in safeguarding gradient updates with minimal impact on model utility. The local explainability module offered actionable insights for HR professionals, with high agreement between explanation methods and strong trust ratings from human evaluators.

Beyond achieving state-of-the-art performance, HFAN-Priv addresses critical real-world concerns around data confidentiality, regulatory compliance, and organizational autonomy. By allowing institutions to collaboratively benefit from shared model intelligence while preserving internal data sovereignty, this framework establishes a practical foundation for ethical AI deployment in human resource management. Future work may extend HFAN-Priv to support asynchronous client participation, handle longitudinal employee trajectories, and integrate reinforcement learning for continuous performance optimization.

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